ConvLSTM (conv_final): a recurrent memory per cell, read out as the maximum over 24 hours

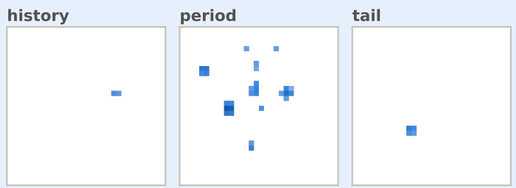
1 Input x (one 12 h period)

24 hourly steps = max of four 15-min slots
6 h history · 12 h period · 6 h tail

7 channels per step and cell
- frp — log(1 + FRP) / log(1 + 1100)
- confidence — confidence / 100
- detected — 1 if detected in the hour
- period — 1 inside the 12 h period
- time — step / 23
- pt — mainland Portugal mask
- persistent — off-season pixel flag

shape 24 × 7 × H × W
training: H = W = 32 (crops)
inference: 177 × 106 (full grid)

example crop (32 × 32 cells): max frp per part



2 ConvLSTM (conv_final): 45,009 parameters

h, c carried to the next hour

Encoder, every hour
Conv 3×3, 7 → 16, ReLU
Conv 3×3, 16 → 16, ReLU
same weights for
all 24 steps
→ 24 × 16 × H × W

ConvLSTM layer 1
16 hidden channels
3×3 gate convolution
hour 1 → 24,
state starts at 0
→ 24 × 16 × H × W

ConvLSTM layer 2
16 hidden channels
reads layer 1's h
per hour
→ 24 × 16 × H × W

Max over time
max of layer-2 h
over all 24 steps
→ 16 × H × W

Inside one ConvLSTM cell, at hour t (one 3×3 convolution, then element-wise updates per cell):

$[i, f, o, g] = \text{Conv3×3}([x_t, h_{t-1}])$ $\quad c_t = \sigma(f) \odot c_{t-1} + \sigma(i) \odot \tanh(g)$ $\quad h_t = \sigma(o) \odot \tanh(c_t)$

Every cell keeps its own memory c; neighbours enter only through the 3×3 convolution, so context spreads about one cell per hour and layer.
The reference ConvLSTM (ref) uses 2 × 32 channels and reads out only the last hour; conv_final reads the maximum, which keeps early hours.

trained with 2 hard negatives per positive

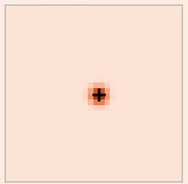
3 Head

Conv 3×3 → 32,
ReLU
Conv 1×1 → 1 logit
sigmoid → p

p = probability that an ICNF fire ≥ 50 ha started in this cell during the 12 h period

4 Target y

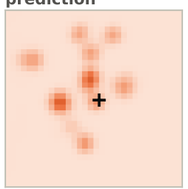
σ = 1 cell



focal loss vs. p

5 Output p

prediction



3 × 3 peaks, p ≥ 0.05
→ sub-pixel points
(lat, lon, score)

example: 2016-08-06
06–18 UTC (validation)
+ = target ignition